Humanized 4-1BB Mouse

Strain Name: C57BL/6-*Tnfrsf9* $^{em1(hTNFRSF9)Smoc}$ Strain Background: C57BL/6 Cat. No. : IT-HU-190077

4-1BB (also known as TNFRSF9 and CD137), a member of the tumor necrosis factor (TNF) receptor family, is mainly expressed on activated T cells. Upon the binding of its agonistic antibodies or its natural ligand, 4-1BB delivers a co-stimulatory signal to T cells, which synergizes with the primary TCR signal to enhance cell proliferation and induce cytokine secretion.

Construction strategy

The humanized 4-1BB mouse model was developed on the C57BL/6 background. A chimeric expression cassette that encodes the extracellular domain of human 4-1BB as well as the transmembrane and intracellular domains of murine 4-1bb was inserted immediately downstream of the start codon of the mouse endogenous 4-1bb gene, followed by a poly(A) element. Thereby, the extracellular domain of the mouse 4-1bb was replaced by its human counterpart while the rest of the mouse gene was retained.

Validation data

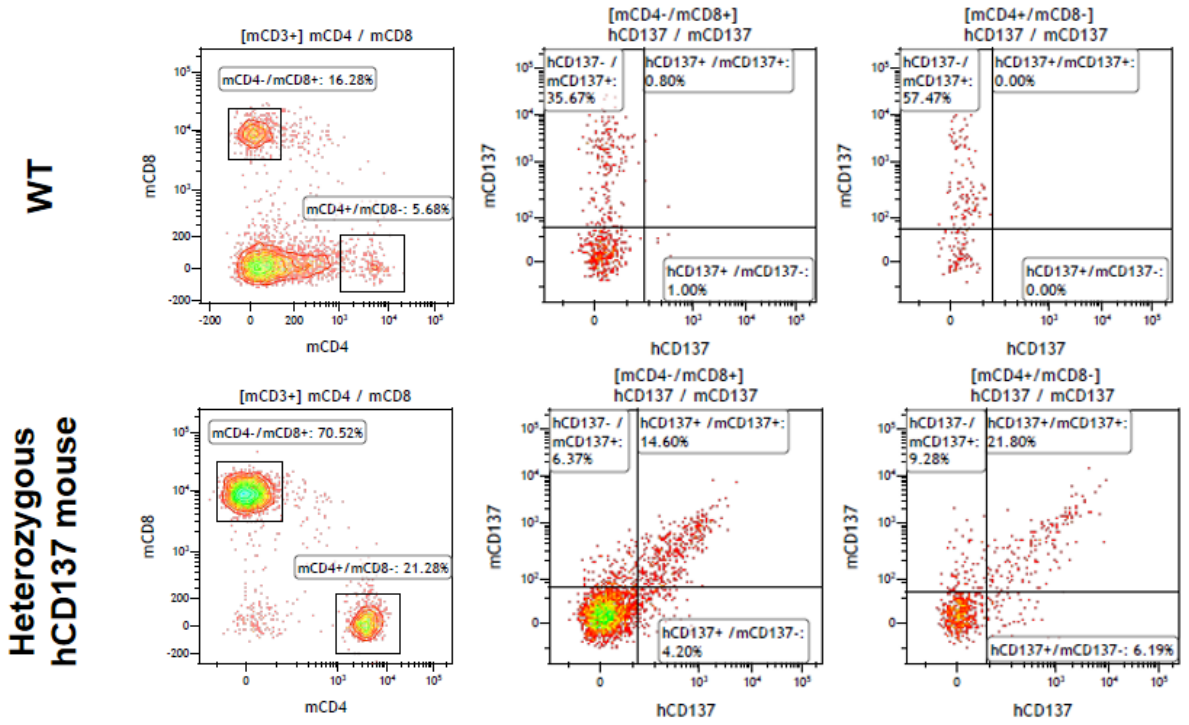


Figure 1. Expression of humanized CD137(4-1BB) in the activated spleen lymphocytes of heterozygous humanized 4-1BB mice is detected by FACS.

The spleen lymphocytes of heterozygous humanized 4-1BB mice were activated by anti-CD3 and anti-CD28 for 48 hours, and then collected for staining. The expression of humanized 4-1BB was detected by FACS. The results showed that the active expression of humanized 4-1BB can be detected in both activated CD4+ and CD8+ T lymphocytes collected from heterozygous humanized 4-1BB mice. (Cooperation with CrownBio)

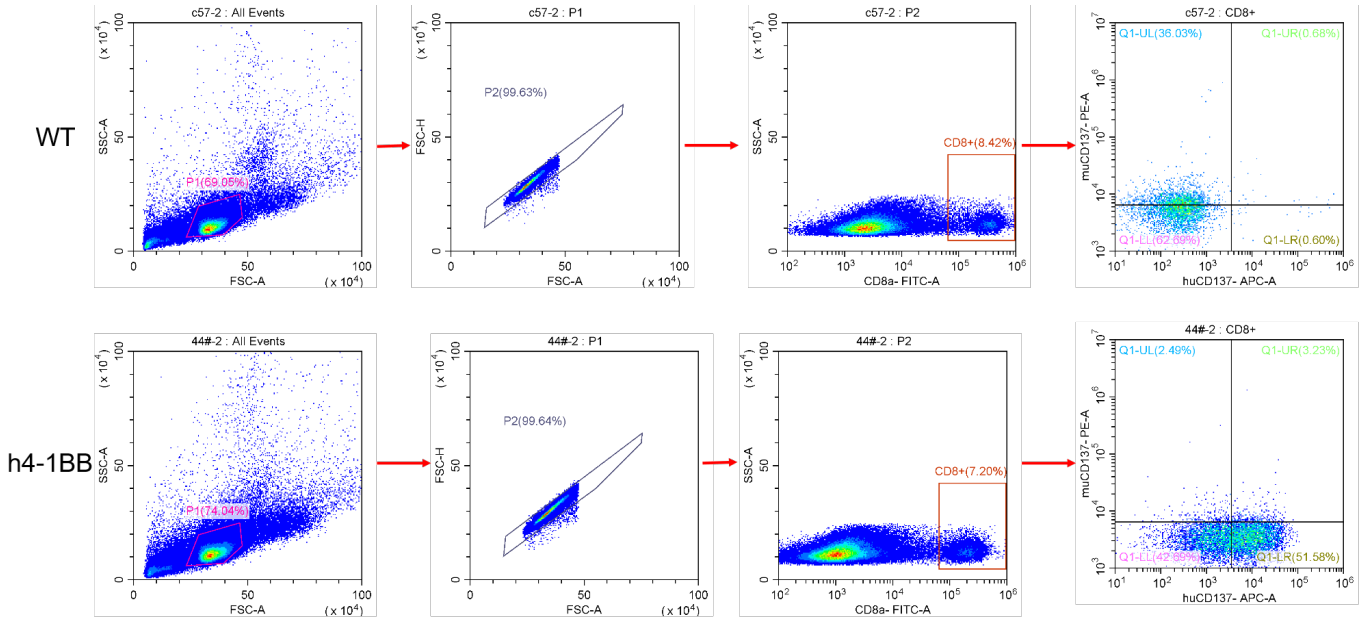
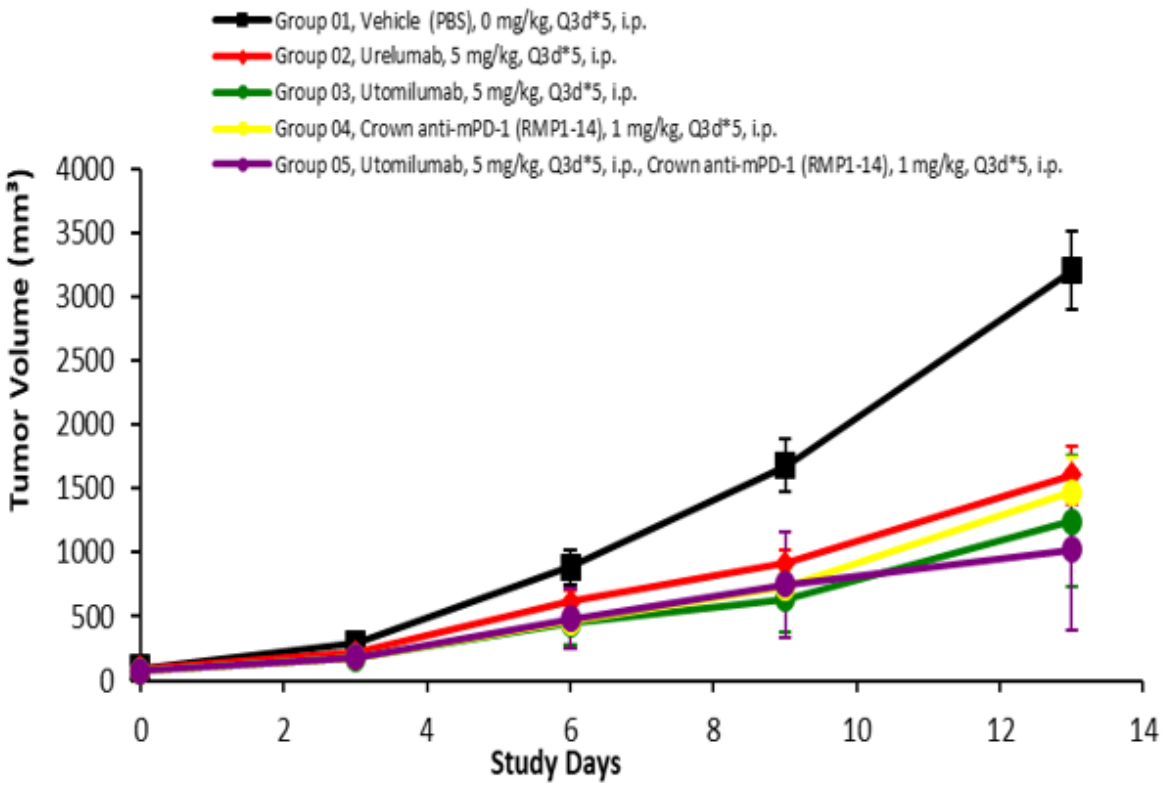


Figure 2. Analysis of human 4-1BB(CD137) expression in the spleen by FACS. The homozygous KI mouse expresses human 4-1BB on the $CD8^+$ T cells.



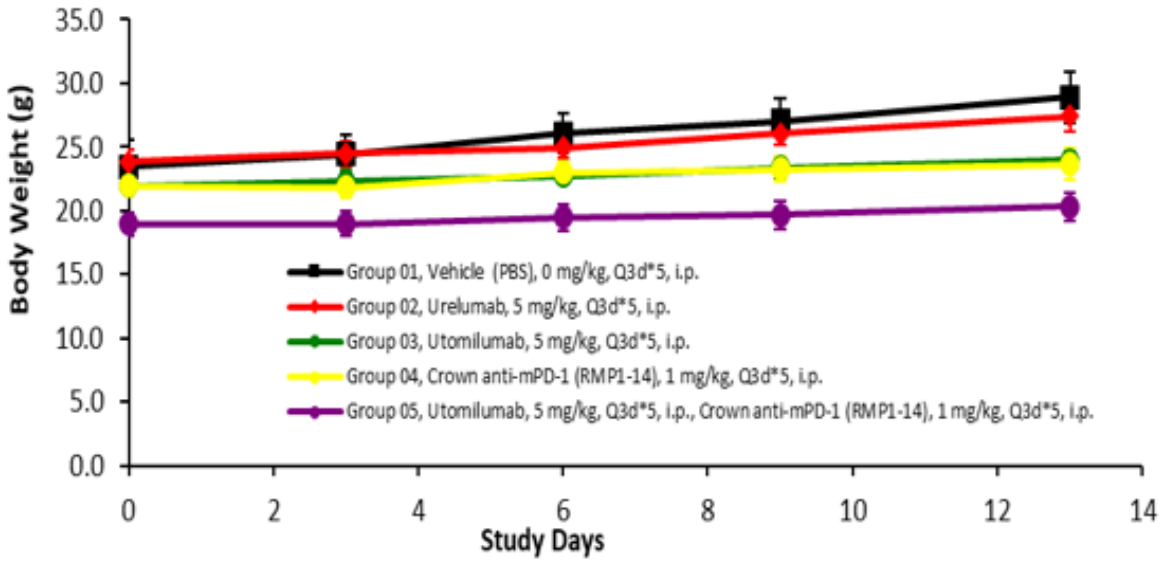


Figure 3. In vivo validation of humanized 4-1BB mice. Humanized 4-1BB mice were inoculated with MC38 colon cancer cells, and were randomly assigned into different groups (n=6) when the tumors grew to a volume of 100 mm3. Human-specifc, 4-1BB antibody (Urelumab or Utomilumab) was given every three days (Q3D) for two weeks (In collaboration with Crownbio).

Immune Checkpoint Humanized Mouse Models

Being recognized as a top scientific breakthrough in 2013, cancer immunotherapy is predicted to be one of the most promising research areas for improving patient outcomes. Although many immunotherapy breakthroughs may still lie ahead, important clinical advances have been made in the past few years for some of the deadliest cancers, reaffirming the potential of immunotherapy for many types of patients.

However, it is worth noting that drug candidates developed to interfere with human proteins may not comparably interact with their murine counterparts. It is therefore critical to develop humanized mouse models to enable in vivo efficacy evaluation of cancer immunotherapies.

Immune Checkpoint Humanized Mouse Models available at ingenious targeting laboratory

4-1BB
CD40
CD47
CD73 (NT5E)
CTLA4 (C57BL/6)
CTLA4 (BALB/c)
KDR
LAG3
OX40
OX40/CTLA4
PD-1 (C57BL/6)
PD-1 (BALB/c)
PD-1/4-1BB
PD-1/CD40
PD-1/CTLA4
PD-1/LAG3
PD-1/OX40
PD-1/PD-L1
PD-1/TIGIT
PD-1/TIM3
PD-L1
PD-L1/CTLA4
PD-L1/LAG3
PD-L1/OX40
PD-L1/TIGIT
SIRPA
SIRPA/CD47
TIGIT
TIM3 (C57BL/6)
TIM3 (BALB/c)
TNFRSF1B
And more to come!

To get to know more about these models, visit our website **www.genetargeting.com**
or contact our scientific experts at **inquiry@genetargeting.com**

About ingenious targeting laboratory

ingenious targeting laboratory (**ingenious**) has been a leading global provider of custom genetically modified mouse, rat, and rabbit models for over 20 years. As one of the very first mouse gene targeting companies, our trusted service is built on two decades' worth of successful animal model creation for investigators, organizations, and companies worldwide. Our models have been published in hundreds of journals including *Science*, *Nature*, and *Cell*, making us one of the most validated and respected production companies in the industry. We are excited to add catalog mouse models to our service repertoire by means of our collaboration with Shanghai Model Organisms Center (SMOC).

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